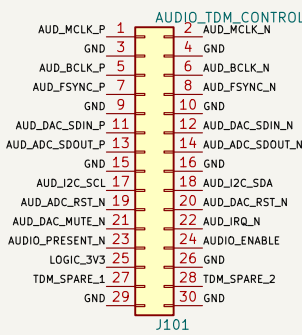
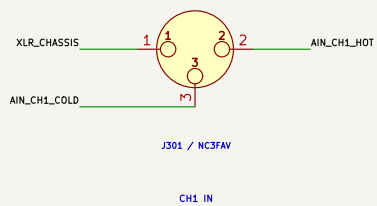
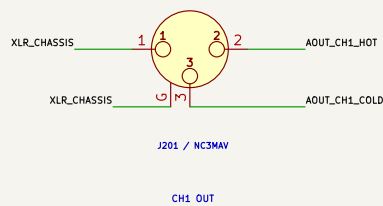


A. PANEL-SUPPORTED BALANCED XLR BANK

B. CARRIER INTERFACES

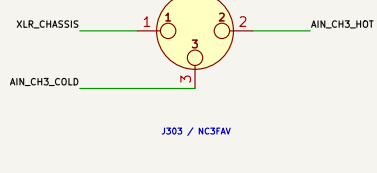
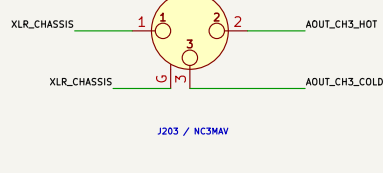


Local filtered conversion produces +/-15 V, AKM 5 V analog and audio 3.3 V.

Mates with CMS-CARRIER J201; 100 ohm differential pairs with interleaved grounds.

C. INPUT SIGNAL CHAIN

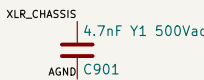
F. XLR SHIELD / CHASSIS BOND



NC3FAV -> chassis/RFI/ESD -> THAT1206 -> level/filter network -> AK5558VN

Eight differential channels; no phantom power; +4 dBu nominal / +24 dBu max.

Detailed A1 sheet captures TDM256 straps, four VREFH feeds, low-noise rails, and the +24 dBu level map.



Every XLR pin 1 and NC3MAY shell G bonds directly to the local chassis plane.

R901 and Y1-rated C901 form the shingle controlled RF/static bond to AGND; verify in EMC test.

No audio return current may use the XLR_CHASSIS copper or panel fasteners.

D. OUTPUT SIGNAL CHAIN

AK4458VN -> reconstruction/gain filter -> OPA165x -> THAT1646 -> RFI/fault network -> NC3MAY

Stable into 600 ohm test load; 10 kohm or higher is the normal operating load.

Hardware mute remains asserted through boot, clock loss, reset and power transfer.

E. DETAILED A1 CAPTURE SHEETS

01 Audio-TDM-Clock: LVDS TDM, protected I2C/control, clock distribution

02 AK5558-ADC: exact pins, TDM256/32-bit I2S straps, references and bypass

03 AK4458-DAC: exact pins, serial control, references and reconstruction contract

04 Audio-Inputs: 8x THAT1206, OPA1652, RFI/ESD and anti-alias networks

05 Audio-Outputs: 8x OPA1652/THAT1646, fail-silent relays and protection

06 Audio-Power: TR120 +/-15 V, 5.5 V pre-rail, dual LT3045, 3.3 V and star

Routing hold: AKM exposed-pad coupons and TQ2 insertion/land-pattern coupon

ENGINEERING CAPTURE – detailed sheets complete; coupon and bench gates remain
Left bank male outputs; right bank female inputs
+4 dBu nominal, +24 dBu maximum, active-balanced XLR stages
AK5558VN ADC and AK4458VN DAC over buffered differential TDM

ProComm
Sheet: /
File: Audio-8x8.kicad_sch

Title: Radxa CM5 ProComm Audio 8x8 – Interface Contract

Size: A2 Date: 2026-08-16 Rev: A1

KiCad E.D.A. 10.0.5-rc1 Id: 1/1